COURSE CODE: CIP-B101

Lab 1A: PC System Fundamentals, Disk Sectors an d Partition Analysis

BY

AKIRA MERCY

C11/26/PCDFIR/17020

INSTRUCTOR: ABDULLAHI UMAR ABBAS

7th November, 2026

1.0 INTRODUCTION

Disk sectors and partitions are fundamental concepts in understanding how compute r storage devices organize and store digital information. This practical assignment fo cuses on the fundamentals of disk structures, sectors organisation and partition anal ysis. The purpose is to develop an understanding of how data is physically and logic ally stored on storage devices. It also aims to introduce the identification and examin ation of disk partitions and their structures. These skills are important in digital foren sics because they help investigators understand storage layouts locate relevant data and examine potential digital evidence.

TOOLS AND COMMAND USED

No.	TOOLS	COMMAND
1	Windows 11pro (Powershell, cmd)	1. Get-Disk Get-Partition Get-Volume 2. Tasklist 3. diskpart list disk select disk 0 list partition 4. systeminfo
2	Linux/Kali VM	**1.** lsblk -o NAME,SIZE,TYPE,FSTYPE,MOUNTPOINT sudo fdisk -l sudo parted -l blkid df –h **2.** lsblk -f blkid df -Th
3	Windows Snipping Tool	For screenshots
4	Calculator	To calculate partition size

Part A - System Identification and Evidence Layer Mapping

In this section, you will identify the major computer system layers and explain how each layer can produce forensic evidence.

A1. Capture basic system specification

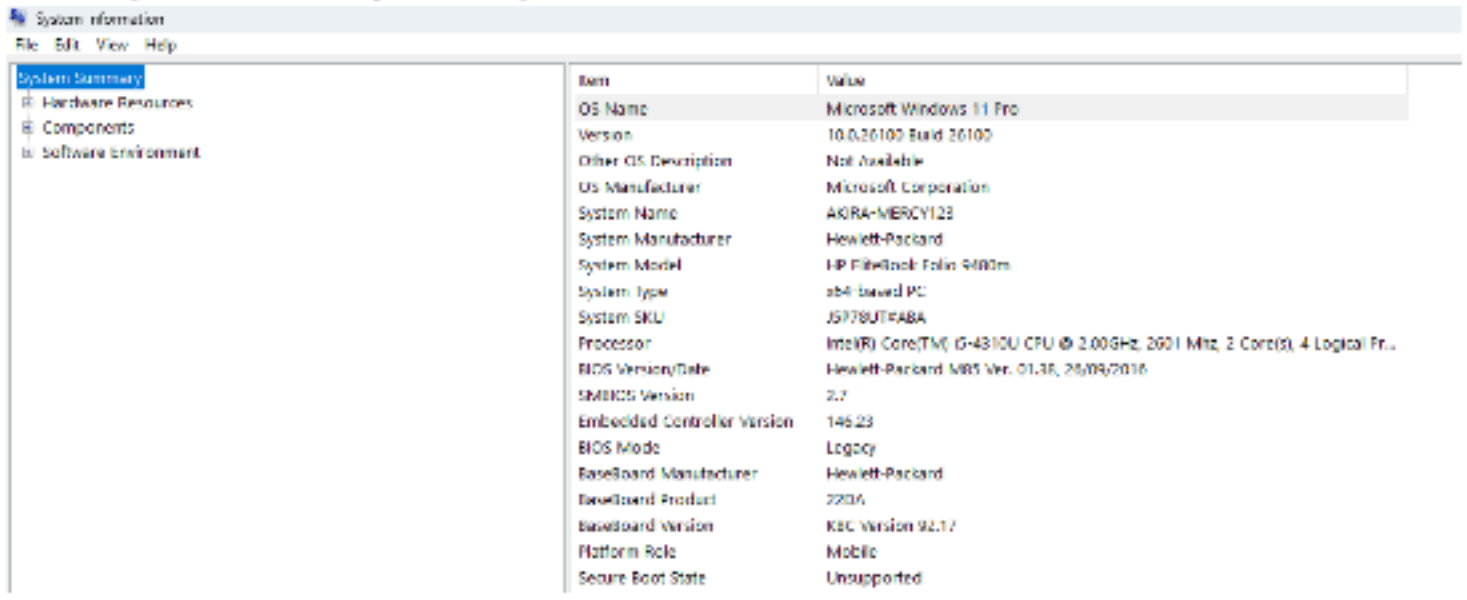


Fig 1: A screenshot showing my system information

```
Command Prompt

Microsoft Windows [Version 10.0.26100.1742]
(c) Microsoft Corporation. All rights reserved.

C:\Users\MERCY>systeminfo

Host Name:                 AKIRA-MERCY123
OS Name:                   Microsoft Windows 11 Pro
OS Version:                10.0.26100 N/A Build 26100
OS Manufacturer:           Microsoft Corporation
OS Configuration:          Standalone Workstation
OS Build Type:             Multiprocessor Free
Registered Owner:          MERCY
Registered Organization:   N/A
Product ID:                00330-80000-00000-AA886
Original Install Date:     02/04/2024, 01:41:11
System Boot Time:          05/11/2026, 14:53:05
System Manufacturer:       Hewlett-Packard
System Model:              HP EliteBook Folio 9480m
System Type:               x64-based PC
Processor(s):              1 Processor(s) Installed.
                           [01]: Intel64 Family 6 Model 69 Stepping 1 GenuineIntel ~2601 Mhz
BIOS Version:              Hewlett-Packard M85 Ver. 01.38, 26/09/2016
Windows Directory:         C:\WINDOWS
System Directory:          C:\WINDOWS\system32
Boot Device:               \Device\HarddiskVolume1
System Locale:             en-us;English (United States)
Input Locale:              en-us;English (United States)
Time Zone:                 (UTC+01:00) West Central Africa
Total Physical Memory:     8,073 MB
Available Physical Memory: 4,446 MB
Virtual Memory: Max Size:  8,585 MB
Virtual Memory: Available: 5,028 MB
Virtual Memory: In Use:    3,557 MB
Page File Location(s):     C:\pagefile.sys
Domain:                    WORKGROUP
```

Fig 2: A screenshot showing my system info on the windows command prompt

A2. Capture storage information

```
Windows PowerShell

PS C:\Users\MERCY> Get-Disk

Number Friendly Name Serial Number                    HealthStatus         OperationalStatus      Total Size Partition
                                                                                                  Style
------ ------------- -------------                    ------------         -----------------      ---------- ----------
0      HGST HTS72...  TF0500Y90THY4L                   Healthy              Online                  465.76 GB MBR


PS C:\Users\MERCY> Get-Partition


   DiskPath: \\?\scsi#disk&ven_hgst&prod_hts725050a7e630#4&29f8f551&0&010000#{53f56307-b6bf-11d0-94f2-00a0c91efb8b}

PartitionNumber  DriveLetter Offset                                        Size Type
---------------  ----------- ------                                        ---- ----
1                            1048576                                     100 MB IFS
2                C           105906176                                   465.03 GB IFS
3                            499432554496                                  642 MB Unknown


PS C:\Users\MERCY> Get-Volume

DriveLetter FriendlyName FileSystemType DriveType HealthStatus OperationalStatus SizeRemaining      Size
----------- ------------ -------------- --------- ------------ ----------------- -------------      ----
                         NTFS           Fixed     Healthy      OK                     69.5 MB    100 MB
C                        NTFS           Fixed     Healthy      OK                    384.9 GB 465.83 GB
                         NTFS           Fixed     Healthy      OK                   112.86 MB    642 MB
```

Fig 3: Screenshot showing my system disk, partition and file system information

A3. Map evidence sources to system layers

Layer / Source	Example Evidence

Layer / Source	Example Evidence
User layer	User accounts and usernames, login and logout activities, recently accessed files, browser history and downloads
Application layer	Application logs ,browser cookies and cache, chat and messaging records and search history
OS layer	System logs and event logs, user account information, file access records, installed program
Disk layer	Deleted files, file system metadata, file timestamps, partition information and unallocated space
Memory	Running processes, encryption keys, open application, malware running in memory

NOTE

Evidence may exist in more than one layer because the layers are interconnected and interact with one another.

Part B - Program, Process and Thread Investigation

This section links the slide concept of program, process, and thread to forensic observation

B1. Observe running processes

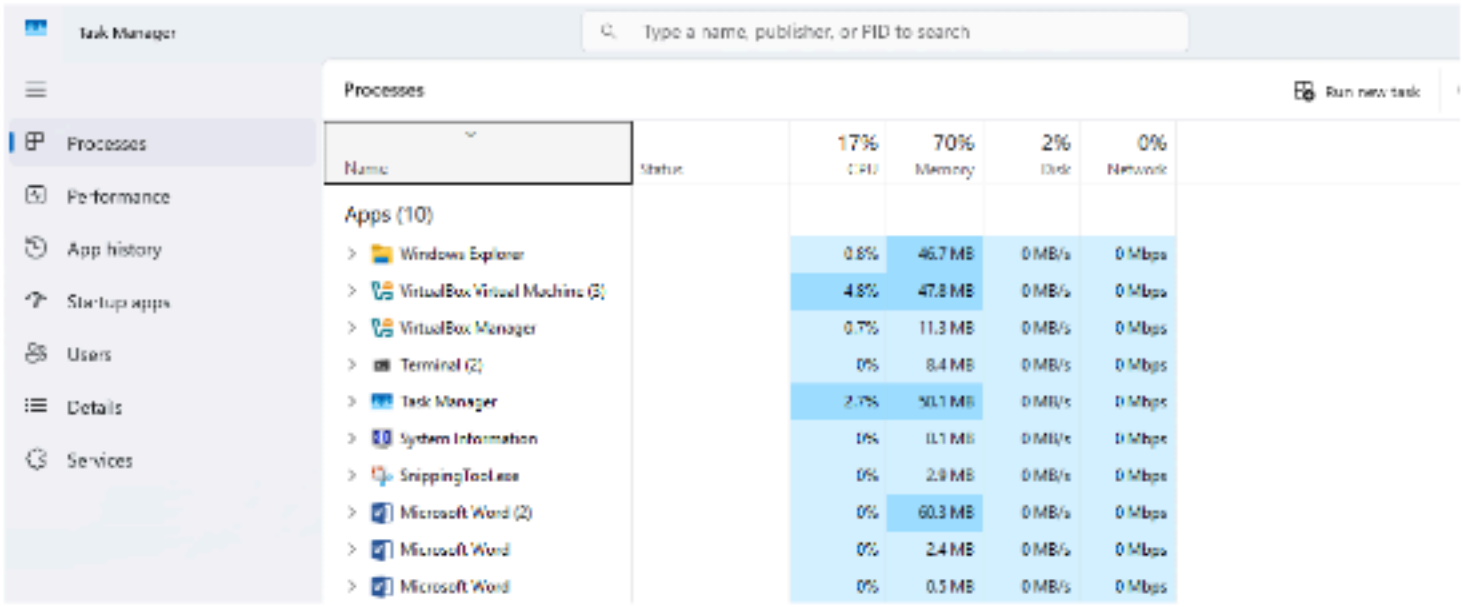


Fig 4: Task manager showing processes running on a system

Prccess name	PID	Likely purpose
WINWORD.EXE	3032	Microsoft Word
Taskmgr.exe	5624	Task Manager
Virtualbox.exe	296	Virtualbox Manager

B2. Explain the difference

PROGRAM: A program is a set of instruction or command given to a computer to perform a spec ific task.

PROCESS: A process is a running instance of a program.

THREAD: Is the smallest unit within a process that allows different tasks within the process to ru n concurrently.

NOTE

Process information is useful during malware analysis and live forensic triage because it helps in vestigators identify what programs are currently running, what they are doing and whether any suspicious activity is running on the system.

Part C - Disk Sector and Disk Size Calculation

C1. Reproduce the slide calculation

Use the values from the slide to calculate the total disk size: sectors per track = 400, heads = 12, cylinders = 17000, and bytes per sector = 512

Total sector = 400 x 12 x 17000 = 81600000 sectors

Total bytes = 81600000 x 512 = 41779200000 bytes

GB = 41779200000/1024/1024/1024 = 38.91GB



Fig 5: Screenshot showing the calculation of partition size

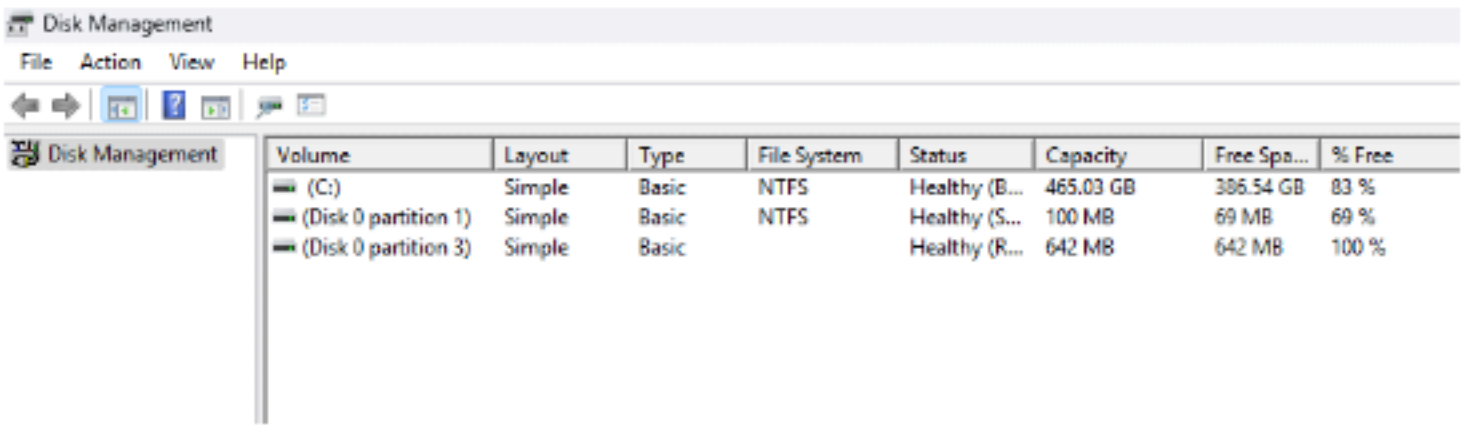


Fig 6: partition table of windows OS

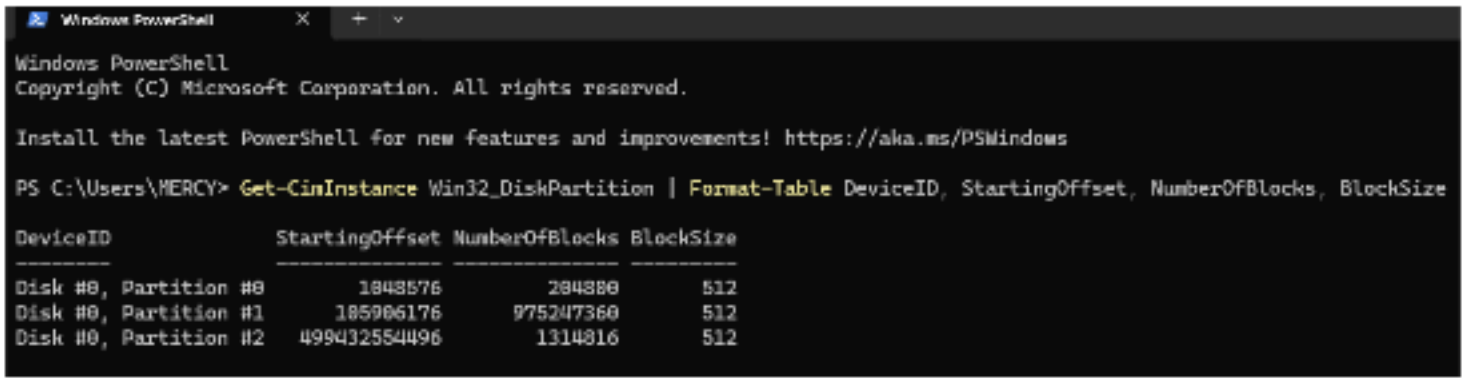
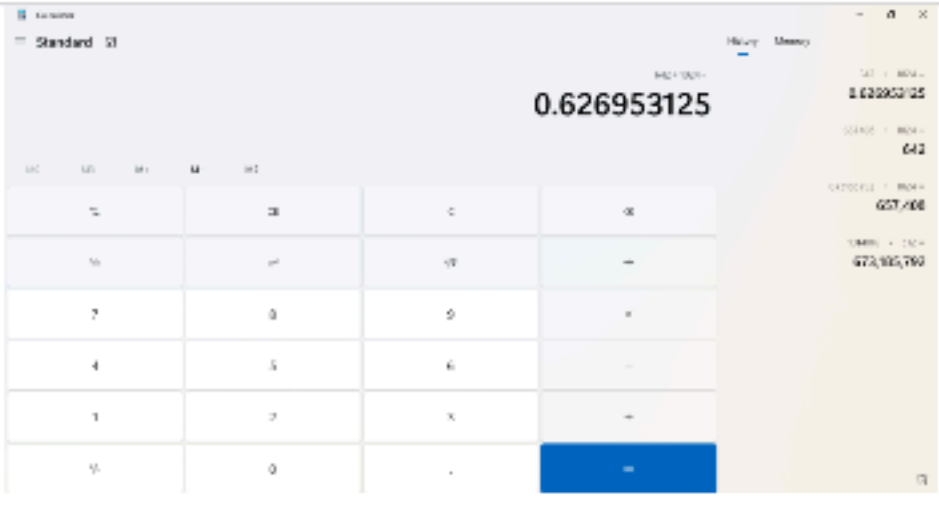


Fig 7: Partition table showing starting offset, Number of blocks, Block Size

. Key formula

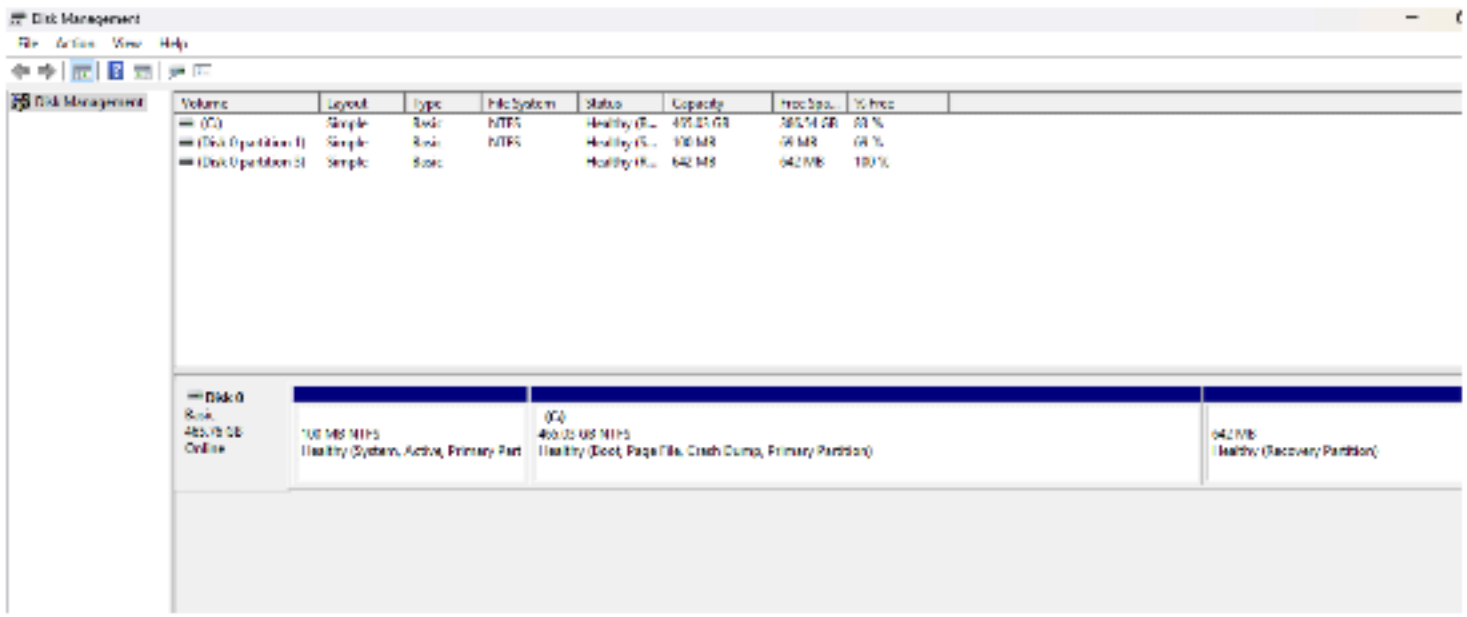
Disk or partition size = number of sectors x bytes per sector. To convert bytes to GB, divide by 1024 x 1024 x 1024.

Partition	Formula	Answer In GB
Partition 1	975247360 x 512 /1024/1024/1024	 465.03GB
Partition 2	1314816 x 512/1024/1024/1024	 0.62GB

Part D - Safe Partition Table Inspection

Choose either Windows or Linux. You may do both for extra practice. You must not modify the disk.

D1. Windows method - Disk Management, PowerShell and DiskPart



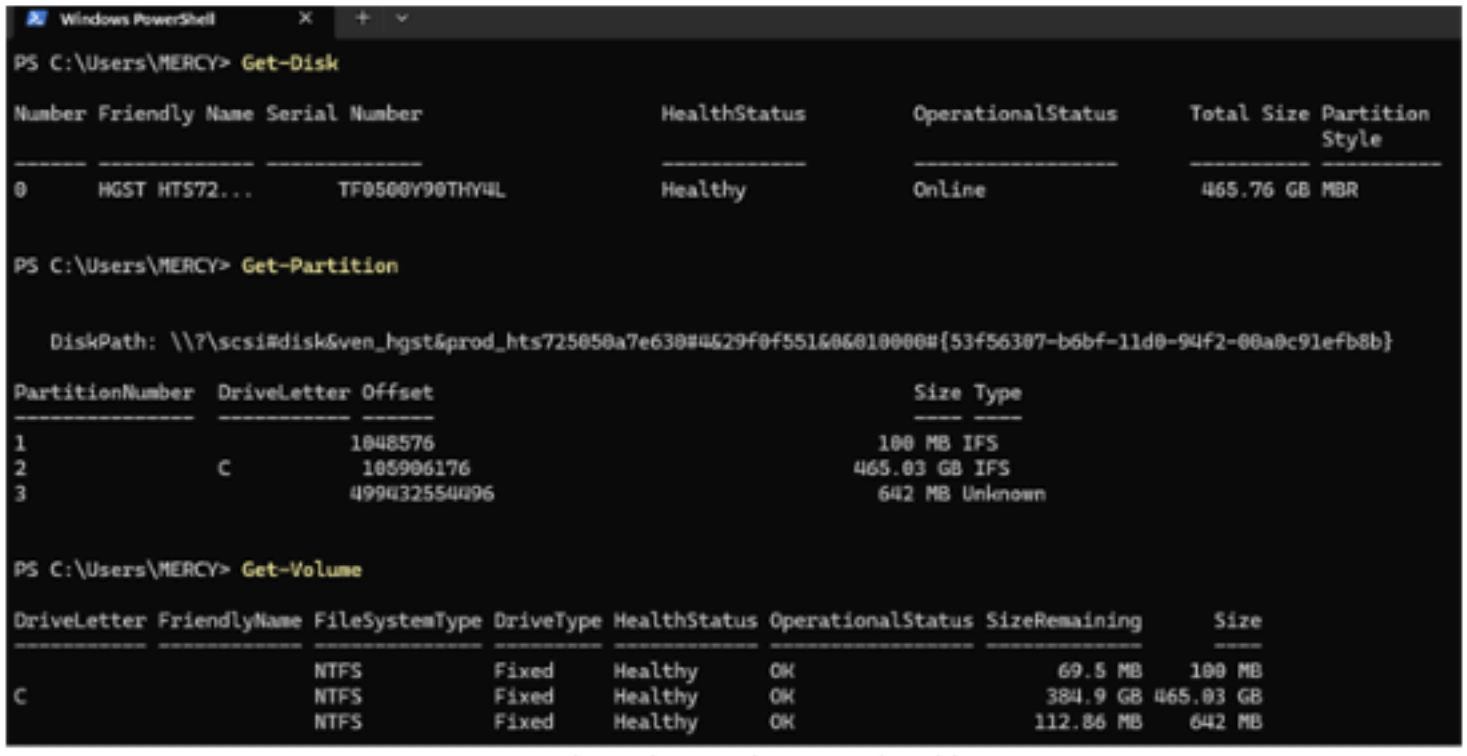


Fig 8: Screenshots showing layout and visible partition

D2. Linux/Kali method - read-only inspection

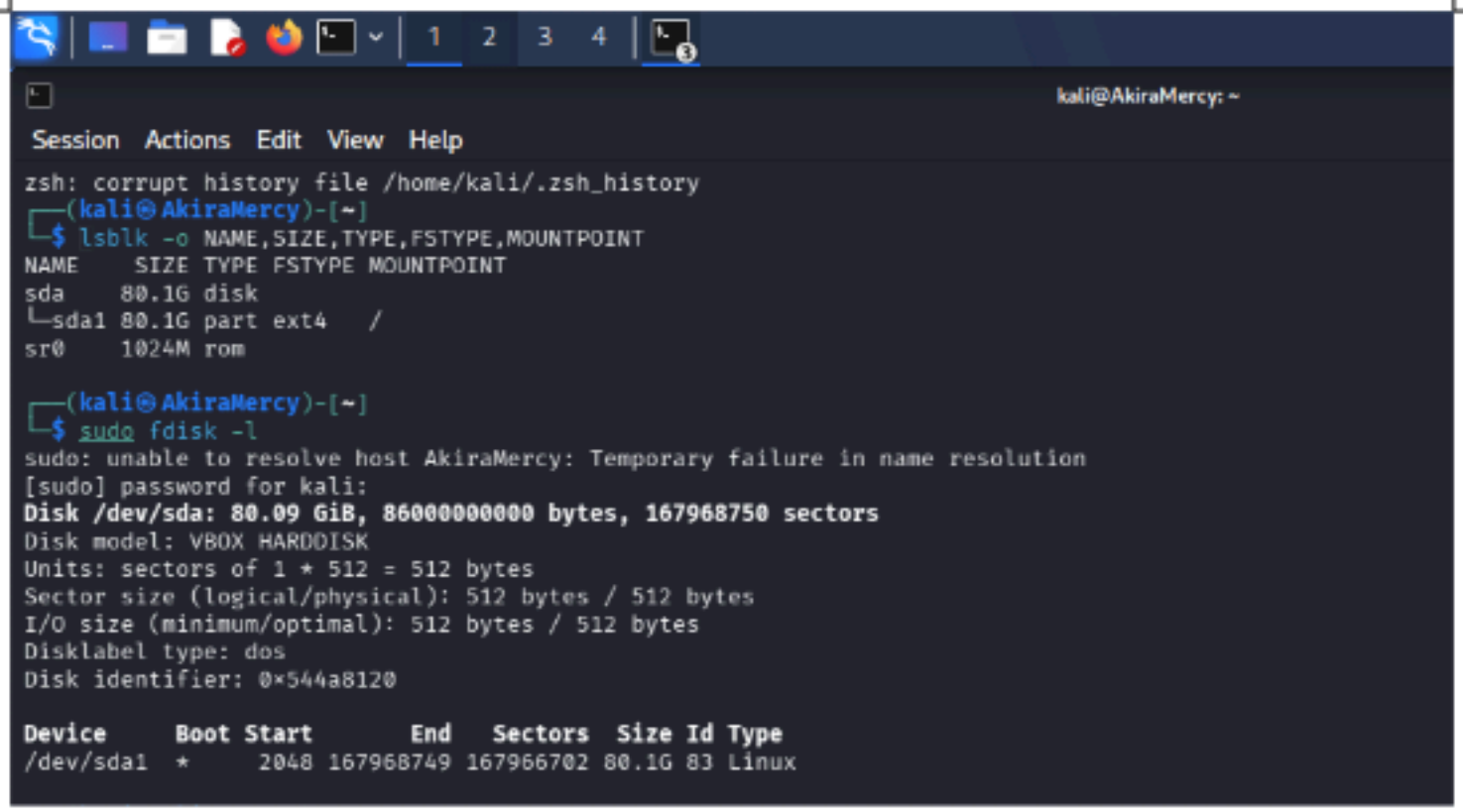


Fig 9: Screenshot showing layout and visible partition

Part E - MBR, GPT and Boot Process Explanation

E1. Explain the boot process

The boot process is a sequence of steps a computer follows when it is switche d on before the Operating system is ready to use.

1. Power on: The computer is switched on and power is supplied to the components.
2. CMOS: stores BIOS/UEFI configuration settings such as boot order, d ate and time.
3. Firmware (BIOS/UEFI): Initializes hardware and performs POST (Powe r-on self-test) to check whether essential components are still workin g.
4. Boot Device Selection: BIOS/UEFI identifies bootable storage device (HDD/SSD).
5. MBR/GBT: The partitioning scheme helps the system identify disk par titions.
6. Bootloader: The system locates and executes the bootloader
7. Kernel loading: The bootloader loads the operating system into the R AM
8. Operating system initialization: The kernel initialize hardware, loads dri vers and starts essential system processes.
9. Operating system takes control: the OS manages the CPU, memory, st orage, processes and other system resources.
10. Login/Desktop: The login screen appears and the computer becomes ready to use.

E2. Compare MBR and GPT

Features	MBR	GPT
Full Meaning	Master Boot Record	GUID Partition Table
Introduction	Older Partitioning scheme	Newer Partitioning scheme
Partition limit	Maximum of four primary partitions	Usually 128 partitions on windows
Disk size	Supports up to 2TB (with 512 bytes sector)	Supports disks larger than 2TB
Boot mode	BIOS(Legacy)	UEFI
Data storage	Stores partition information in the first sector	Stores partition information in headers and partition entries
Backup	No built-in backup of partition table	Keeps a backup of partition information
Reliability	Less reliable if the MBR is damaged	More reliable due to backup and CRC check

Part F - File System Identification and Interpretation

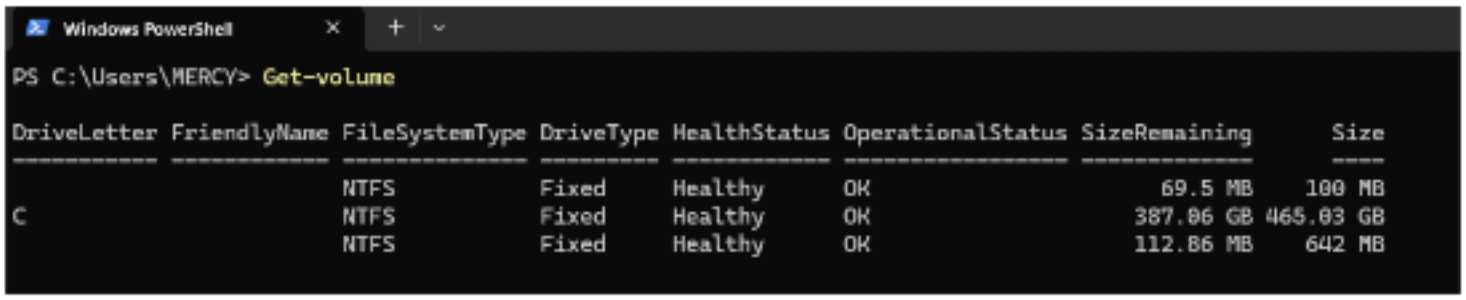


Fig 10: Screenshot showing Volumes on windows system using NTFS file system

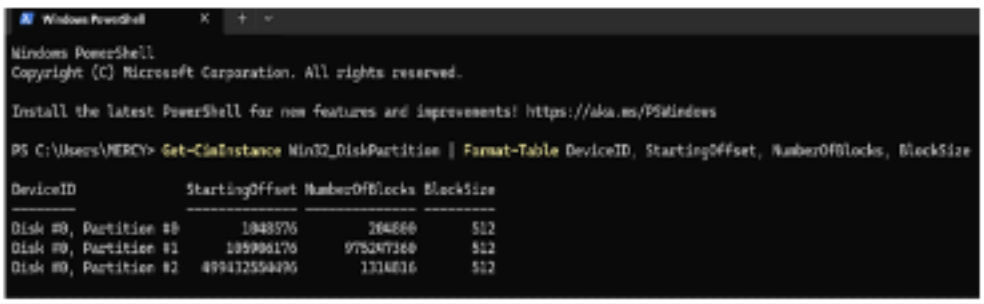
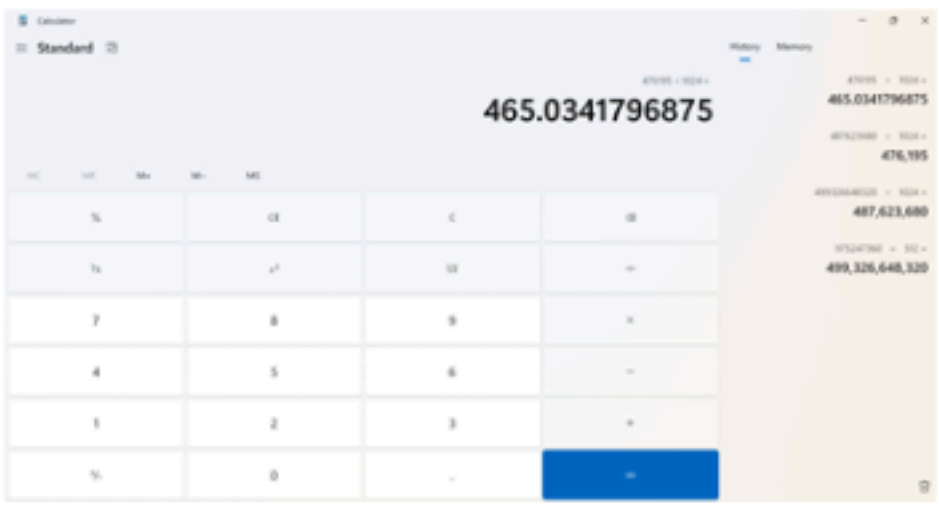


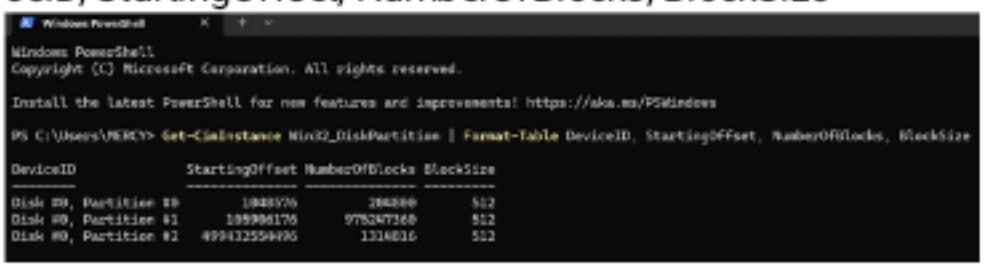
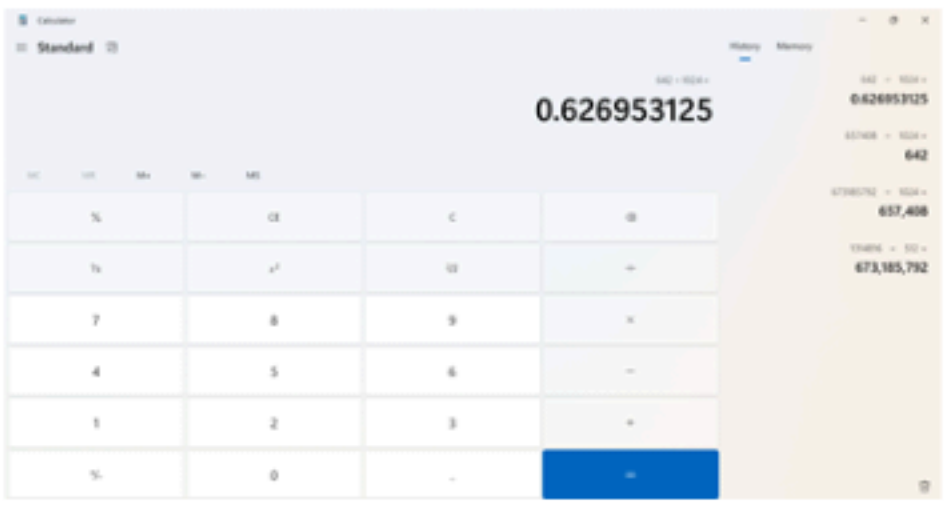
Fig 11: Screenshot showing Linux system using ext4 file system

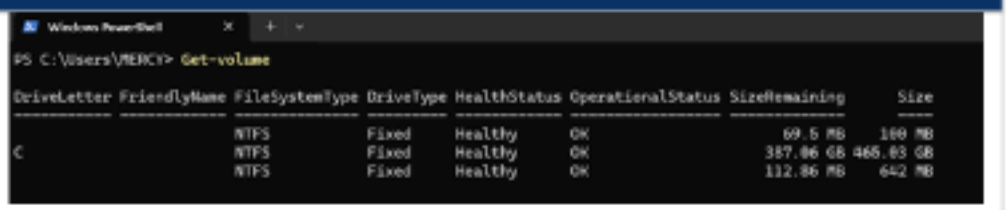
NOTE

File system is important in Digital forensics because it helps investigators unde rstand how data is stored, organized, accessed, modified and deleted on a stor age device.

Part G - Final Practical Assignment

	Required Evidence	Student Notes / Answer
Identify partition 1	Screenshot of partition table and command output	COMMAND USED Get-CimInstance win32_Diskpartition | Format-Table DeviceID, StartingOffset, NumberOfBlocks, BlockSize 
Record partition 1 sector count	Start sector, end sector or sector count	975247360
Calculate partition 1 size	Formula and result in GB	Partition size = Sector count x Bytes size /1024/1024/1024 =975247360 x 512/1024/1024/1024 =465.03GB 

	Required Evidence	Student Notes / Answer
Identify partition 2	Screenshot of partition table and command output	COMMAND USED Get-CimInstance win32_Diskpartition | Format-Table DeviceID, StartingOffset, NumberOfBlocks, BlockSize 
Record partition 2 sector count	Start sector, end sector or sector count	1314816
Calculate partition 2 size	Formula and result in GB	Partition size = Sector count x Bytes size /1024/1024/1024 = 1314816 X 412/1024/1024/1024 =0.62GB 

	Required Evidence	Student Notes / Answer
Compare with OS-reported size	Disk Management, Get-Volume, lsblk, or df output	
Explain difference if any	Short explanation of rounding, reserved space, file system overhead, or GiB/GB differences	1. Rounding Rounding occurs when a calculated value is approximated to a particular number of decimal places or whole numbers. 2. Reserved Space Reserved space is the portion of a disk that is set aside by the operating system or file system for system operations and other important functions. 3. File System Overhead This refers to the storage space used by the file system to manage and organise files rather than store actual user data. 4. GiB/GB Difference Both measure storage capacity, but they use different measurement systems: - GB (Gigabyte): Based on decimal, where 1 GB = 1,000,000,000 bytes. - GiB (Gibibyte): Based on binary, where 1 GiB = 1,073,741,824 bytes.

Conclusion

During this lesson, I learned about hard disk components, including platters, tracks, sectors, cylinders and read and write heads, as well as how to calcul ate disk partition sizes. I also learned about file systems, BIOS/UEFI and th e various stages involved in the computer booting process. These concepts are important to Digital Forensics because they provide an understanding o f how data is stored, organised, accessed and retrieved on a computer. The y also help forensic investigators identify storage locations, recover deleted data, examine file metadata, understand system startup activities and pres erve digital evidence accurately during investigations.